

Photobiomodulation and Mental Health Outcomes: An Evidence Synthesis Dashboard

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Photobiomodulation (PBMt), also known as red light therapy (RLT), uses red and near-infrared light to stimulate tissue repair, reduce inflammation, and relieve pain. Non-thermal, non-invasive, and generally safe, PBMt works at the cellular level by acting on mitochondria to boost energy production (Hamblin, 2017). What began as a public wellness trend has since expanded into clinical applications across industries. Mental health applications have gained particular traction, yielding promising early results. This project maps that research landscape to make it easier to interpret, organized around three questions: Which mental health conditions have the most published research, and how do outcomes vary across them? Who is being studied, and does population type or age shape which conditions receive research attention? How rigorous is the evidence base, and do stronger study designs produce different outcomes? Together, these questions frame an examination of which outcomes are most studied and where the evidence appears strongest, weakest, or most mixed.

Dataset Composition

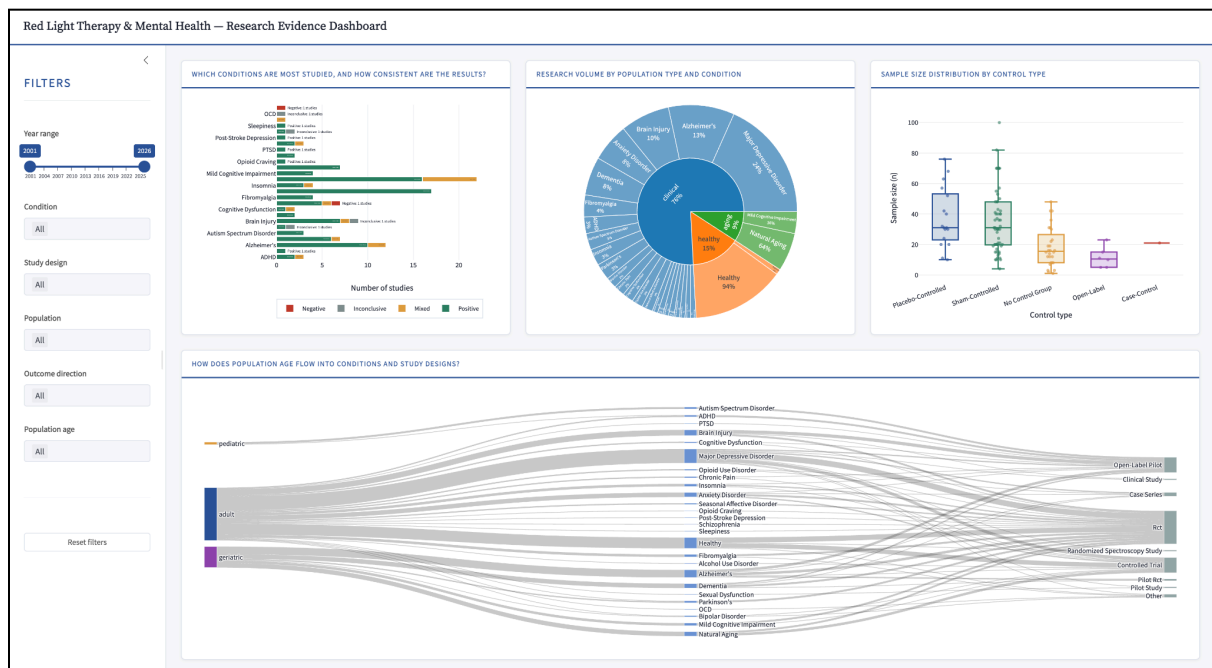
Data were collected using the PICO framework (Population, Intervention, Comparison, Outcome) as a structured lens for determining study and variable eligibility (Schardt et al., 2007). Studies were sourced from PubMed, ScienceDirect, SAGE Journals, and other trusted academic databases. The 102 retrieved studies were processed through a cleaning pipeline in R that standardized formatting and delimiters, applied regex matching to consolidate label variants, and expanded multi-condition and multi-outcome entries into separate rows so each study contributed independently to every condition and outcome it examined. This produced a final dataset of 227 observations across the original 102 studies, with each row representing a unique study-condition-outcome combination. The resultant variables are presented in Table 1.

Table 1

Type	Column
string	author, source, study_id
integer	sample, session_count, year
numeric	session_duration_min, wavelength_nm
categorical	application, blinding, condition, control_type, outcome_cat, outcome_direction, population_age, population_type, publisher, study_design

Methodology and Approach

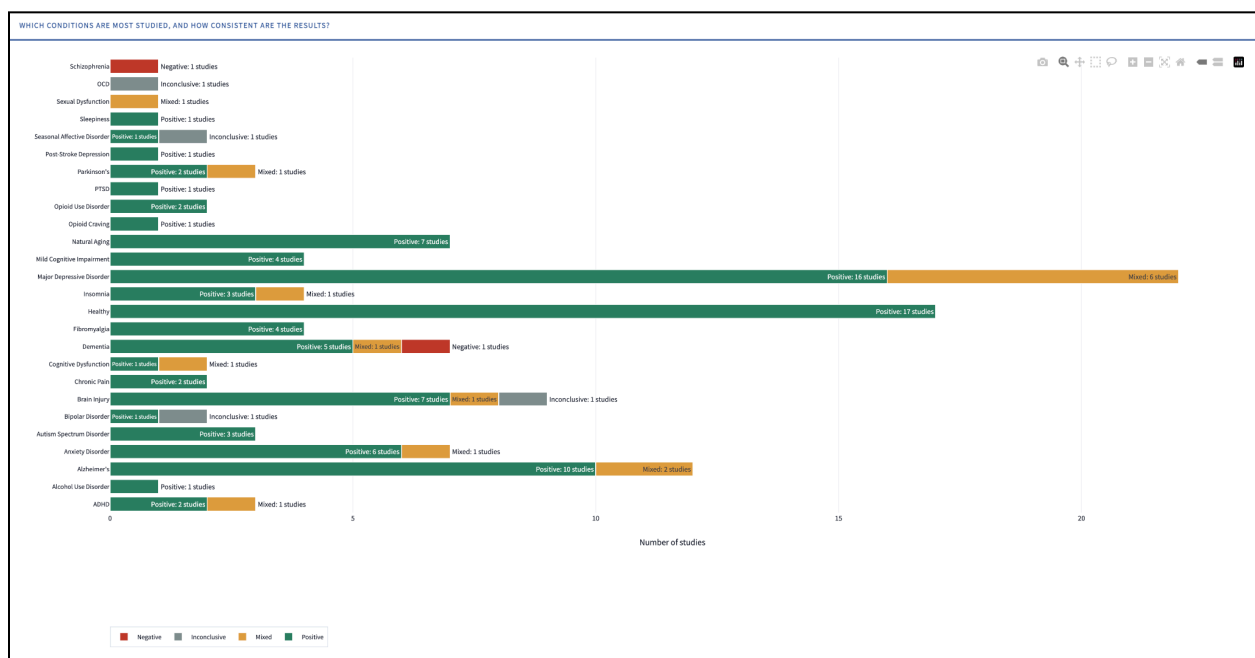
The dashboard is intended to summarize and visualize study characteristics and outcomes from vetted RLT research. Audience matters here: a dashboard designed for the general public, policymakers, and medical researchers would each look very different. We targeted researchers seeking to understand the scope and findings of published studies, positioned between general interest and deep clinical detail. The goal was to clearly communicate the what, remaining informative and accessible without overwhelming the audience with clinical specifics.

Figure 1

The dashboard was built in R using Shiny and Plotly to incorporate interactive plots into a cohesive UI/UX (Figure 1). It is composed of four charts, all filterable by year, condition, study design, population type, outcome direction, and population age. Each chart responds to the selected filters and includes its own interactive elements to surface additional contextual detail on demand. This balance of information and interactivity lets the audience quickly build an overall picture while retaining the ability to drill into specific populations, outcomes, or study designs to examine resultant findings, or the absence of them.

Visualization Selection

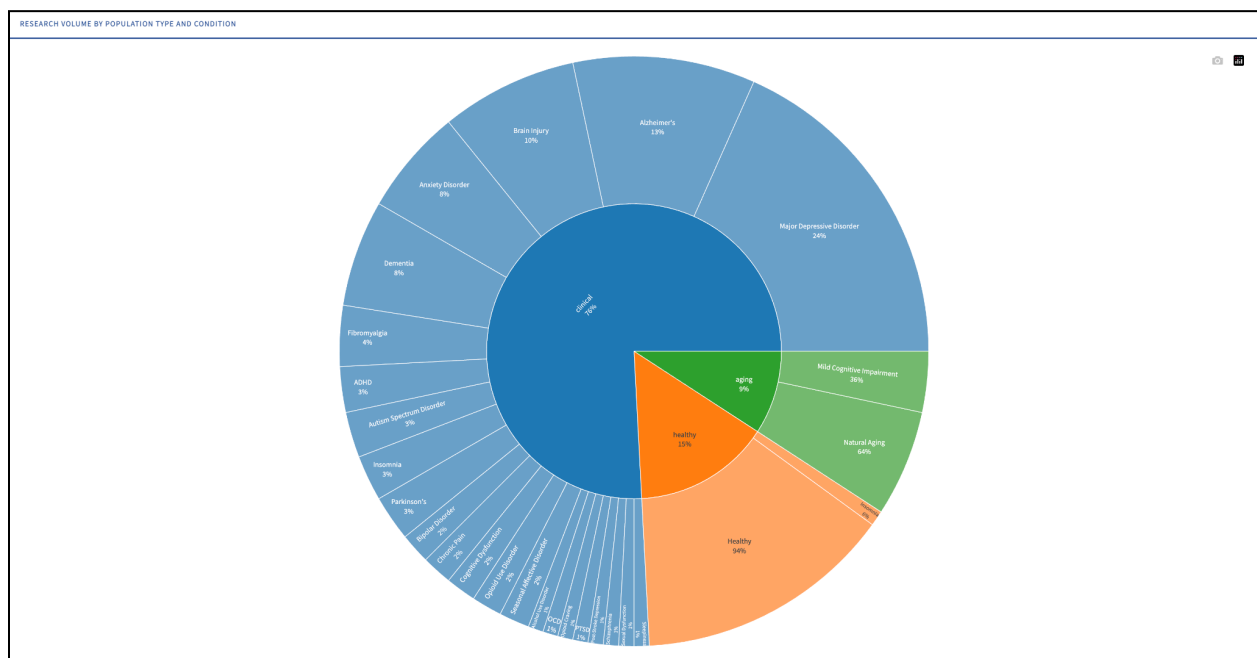
Figure 2



The stacked bar chart (Figure 2) serves as the dashboard's anchor visualization, with each bar representing a condition, bar length encoding total study count, and color indicating outcome direction. This design follows the overview-first principle described by *A Tour Through the Visualization Zoo* as central to effective exploratory visualization (Heer et al., 2010). The dual encoding of quantity through length and composition through color aligns with *Graphical*

perception: Theory, experimentation, and application to the development of graphical methods, as both channels are preattentive and allow viewers to compare conditions without additional cognitive load (Cleveland & McGill, 1984). This makes distributional patterns immediately apparent: MDD dominates in study volume but carries a notable proportion of mixed outcomes, while conditions like Fibromyalgia trend entirely positive but rest on a considerably smaller evidence base.

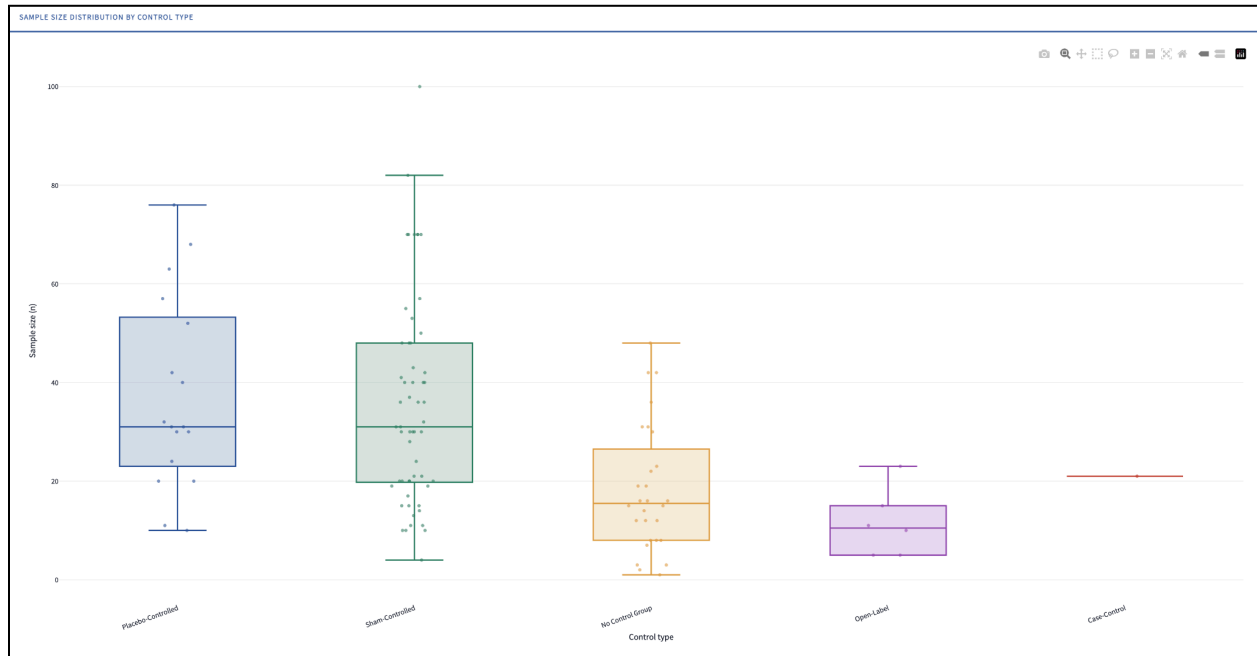
Figure 3



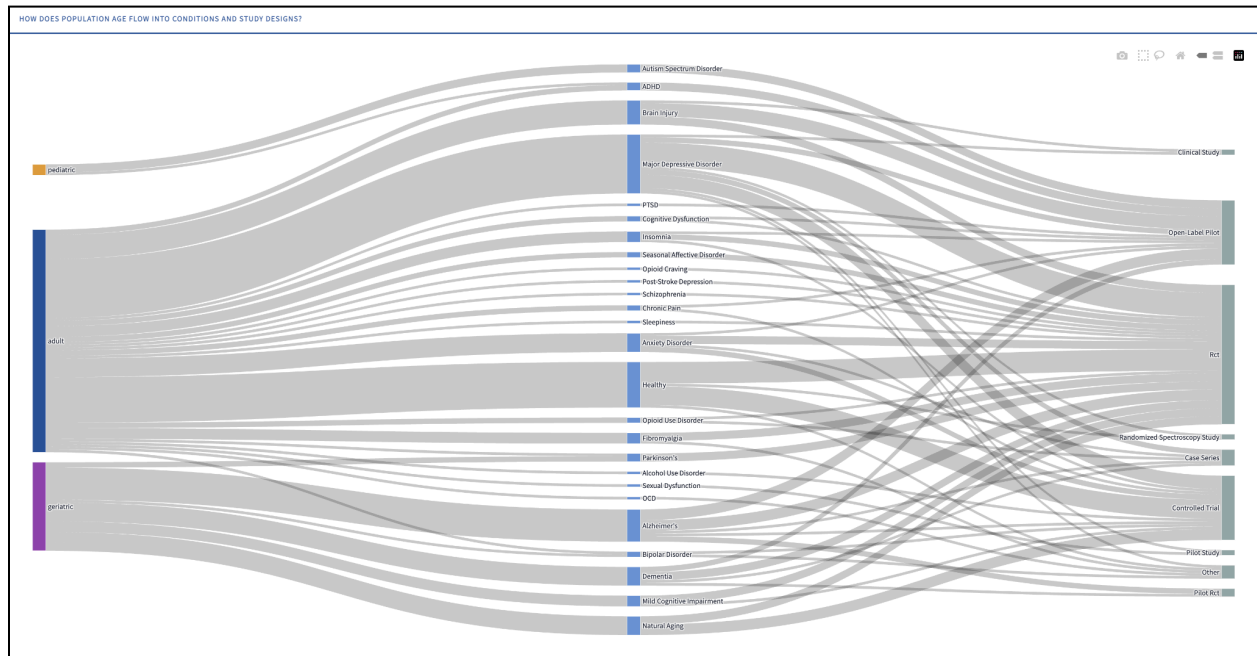
The sunburst plot (Figure 3) was selected to display population and condition distribution across multiple dimensions within a single visual. Rather than a standard pie chart, the sunburst captures parts of a whole while preserving hierarchical structure: the inner ring displays population type (clinical, healthy, and aging) and the outer ring breaks down each condition within those groups. This gives viewers an intuitive sense of both proportion and distribution without requiring navigation across separate charts. As established in visualization theory and reinforced in course material, visuals that minimize interpretive steps reduce cognitive load,

making the sunburst a natural fit for communicating layered categorical relationships efficiently (Leach, 2026a).

Figure 4



The boxplot (Figure 4) displays sample size distribution by control type, communicating spread and central tendency efficiently, fitting for a research audience already familiar with its format. Jittered interactive points are overlaid on each box, allowing users to hover for source and sample size details on demand. This follows the principle of tasteful interactivity discussed in Lecture 20: contextual depth is available without cluttering the static visual, surfacing detail only when the viewer seeks it (Leach, 2026b).

Figure 5

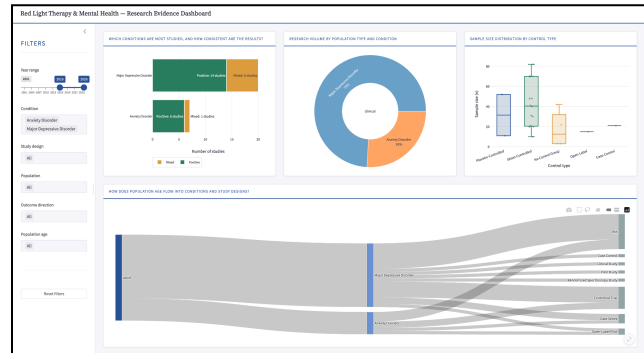
The Sankey diagram (Figure 5) traces flow across three variables simultaneously: population age into conditions into study designs, with flow width encoding study count. Heer et al. identify Sankey diagrams as particularly well-suited for showing how quantities move through a multi-stage system, which maps directly onto the research pipeline structure of this dataset (Heer et al., 2010). The visualization makes several patterns immediately legible: adult populations dominate across conditions, geriatric populations cluster around Alzheimer's and Dementia, and randomized controlled trials appear far more frequently than open-label pilots regardless of condition. This last pattern is among the more substantive findings of the project, speaking directly to the overall rigor of the evidence base.

Research Applications

A psychiatrist specializing in MDD and Anxiety wants a snapshot of the last decade of research across those two conditions (Figure 6). Filtering by condition and dragging the year range to 2016 through 2026, the dashboard immediately narrows the view. The Sankey diagram

shows all flows originating from a single adult node, confirming that no geriatric or pediatric research exists for either condition in this period. MDD fans out across a wide spread of study designs, including RCTs and controlled trials, while Anxiety's flows are noticeably thinner and lean heavily toward open-label pilots, making the evidence gap between the two conditions visually immediate without requiring any additional navigation.

Figure 6



A geriatric care researcher seeking to understand which conditions are being studied in older adult populations follows a similar process (Figure 7). Filtering by population age to geriatrics narrows the view to six conditions. The sunburst chart reveals that 67% of geriatric research falls within clinical populations and 33% under general aging, with Alzheimer's alone accounting for 55% of the clinical slice. With a handful of interactions, the researcher has a clear picture of where the evidence is concentrated and where the gaps remain.

Figure 7



Discussion

The findings of this project surface several patterns that speak directly to the state of PBMt research as a field. Major Depressive Disorder emerged as the most studied condition by a considerable margin, yet the presence of mixed outcomes within that evidence base signals that the field has not yet reached consensus even for its most researched application. Conditions like

Dementia and Anxiety Disorder show nearly uniform positive results, but those findings rest on thin evidence bases dominated by open-label pilots and case series rather than predominantly randomized controlled trials, a distinction that carries significant implications for how those results should be interpreted.

The Sankey diagram makes this structural problem apparent at a glance: regardless of condition, weaker study designs outnumber rigorous ones, and the gap is particularly pronounced for conditions outside the cognitive and depressive disorder categories. The sample size distribution reinforces this, with median samples clustering well below the threshold typically required for adequately powered clinical inference. Taken together, these patterns suggest that the field's broadly positive publication record is at least partly a result of study design choices rather than a reliable signal of therapeutic efficacy. Publication bias is a plausible and likely contributor as well, 82.4% positive outcomes across 102 studies is a statistic that signals bias.

Conclusion

This project set out to map the research landscape of PBMt and mental health outcomes, and the dashboard accomplishes that goal across four interconnected visualizations. The evidence base is real, growing, and concentrated in a handful of conditions, but it is also methodologically uneven in ways that are not apparent from headline outcome rates alone. The interactive filters allow researchers to move beyond the aggregate picture and interrogate specific conditions, populations, and study designs, surfacing gaps and inconsistencies that a static summary would obscure. Future work should prioritize expanding the dataset through automated PubMed integration, incorporating effect sizes and funding source metadata, and building a secondary dashboard page focused on dosing parameters for researchers designing new protocols. The broader takeaway is not to discredit red light therapy as a legitimate intervention

for mental health, but to reveal that the current evidence is not yet strong enough to draw firm conclusions. Making that distinction is precisely what this project was designed to do.

References

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